CEDT Software Engineering semester 2/2566

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Lab Sheet – Software Project Management (Project Estimation, WBS, Gantt Chart, Risk Mgt)

Part A Size, Effort, and Cost Estimation [Estimated time: 1 hour]

1. A web admin uses data analytic software to collect the number of user interactions with the web site (i.e. hits) and the number of user sessions (i.e. user sessions) in a user activity database on a daily basis. At the end of the month, the web admin displays a report to see the activity level by day of the week. Calculate the function point of this function of the software.

Activity Level by Day of the Week

Display

Activity Level by Day of the Week

	Day	Hits	% of Total Hits	User Sessions
1	Sun	1004	8.73%	111
2	Mon	1887	16.41%	201
3	Tue	1547	13.45%	177
4	Wed	1975	17.17%	195
5	Thu	1591	13.83%	191
6	Fri	2209	19.21%	200
7	Sat	1286	11.18%	121
	Total Weekdays	9209	80.08%	964
	Total Weekend	2290	19.91%	232

Which function (EI, EO, EQ, ILF, or EIF)?	EO
What is the number of DETs and what are they?	11 DETs 1. Display (input) 2. Day (output) 3. Hits (output, derived) 4. % of Total Hits (output, derived) 5. User Sessions (output, derived) 6. Total Weekdays – Hits (output, derived) 7. Total Weekdays - % of Total Hits (output, derived) 8. Total Weekdays – User Sessions (output, derived) 9. Total Weekend – Hits (output, derived) 10. Total Weekend - % of Total Hits (output, derived) 11. Total Weekend – User Sessions (output, derived)
What is the number of FTRs and what are they?	1 FTR 1. User activity database (ILF)
What is the number of RETs and what are they?	-
Rating (Low, Average, or High)?	Low
Value?	4

2. The US Census Bureau hosts a web application that allows users to click on a location on the map (e.g. location of Kansas City) to see some information about that location (i.e. city, state, population and coordinate). The US Census Bureau maintains its own population database and obtains the coordinate of the state from the Surveying and Mapping Section of the US Army [กรมแผนที่ทหาร]. Calculate the function point of this function of the application.

Kansas City, Missouri: Population 435,146: Location: 39.1 N, 94.5 W

Which function (EI, EO, EQ, ILF, or EIF)?	EQ
What is the number of DETs and what are they?	5 DETs 1. Click on map location (input) 2. City (output) 3. State (output) 4. Population (output) 5. Coordinate (output)
What is the number of FTRs and what are they?	2 FTRs 1. US Census Bureau population database (ILF) 2. Surveying and Mapping Section of the US Army database (EIF)
What is the number of RETs and what are they?	-
Rating (Low, Average, or High)?	Low
Value?	3

3. The Mor Prom [หมอพร้อม] mobile application originally provides the Covid vaccination service. Users can register to use this service and the registation data will be used by the Ministry of Public Health to monitor the vaccination status of the country. Calculate the function point of this function of the application.

Which function (EI, EO, EQ, ILF, or EIF)?	EI
What is the number of DETs and what are they?	15 DETs 1. คำนำหน้าชื่อ (input, data) 2. ชื่อ (input, data) 3. นามสกุล (input, data) 4. เลขประจำตัวประชาชน (input, data) 5. วันที่เกิด (input, data) 6. เดือนที่เกิด (input, data) 7. ปีที่เกิด (input, data) 8. เบอร์โทรศัพท์ (input, data) 9. เพศ (input, data) 10. จังหวัด (input, data) 11. อำเภอ (input, data) 12. ตำบล (input, data)

In fact, these should be counted as 1 (birthdate)

	13. รหัสไปรษณีย์ (input, data) 14. เลือกโรงพยาบาล (input, data) 15. ปุ่มบันทึก (input, control)
What is the number of FTRs and what are they?	1 FTR 1. Mor Prom database (ILF)
What is the number of RETs and what are they?	-
Rating (Low, Average, or High)?	Low
Value?	3

4. Suppose the function points of all Mor Prom functions have been calculated and the total unadjusted function point for the application is 100. If the application is written in JavaScript, what is the size of this application (in KLOC)? Show your calculation.

Gearing factor is 47 LOC/FP (100*47)/1000 = 4.7 KLOC

5. Using the COCOMO II model and your answer in question 4, what is the project effort in person-month (PM) for the Mor Prom project, provided that alll scale factors are rated as Extra High (i.e. 0.00) and all effort multipliers are rated as Nominal (i.e. 1)? Show your calculation.

E = 0.91+(0.01*0) = 0.91 Project effort PM = $2.94*4.7^{0.91}*1$ = 12.02144 ~ 12 person-months

6. Suppose the Mor Prom project uses the agile methodology (Scrum). There are different types of activities that incur labor costs at different rates. The project effort PM (i.e. your answer in question 5) is distributed among different project activities and different sprint activities as shown below.

Project Activity	% of Project Effort PM		
Management	10%		
Analysis	10%		
Sprint 1	40%	**Sprint Activity**	**% of Sprint Effort**
		Analysis	5
		Design	15
		Implementation	50
		Testing	20
		Management	10
Sprint 2	40%	**Sprint Activity**	**% of Sprint Effort**
		Analysis	5
		Design	15
		Implementation	50
		Testing	20
		Management	10

In the table below, the average activity labor cost for each type of project activities is given. Show how to calculate the % of project effort of each project activity, activity effort in person-months, activity labor cost, and total project labor cost.

Project Activity	% of Project Effort PM	Activity Effort (Person-Months)	Average Activity Labor Cost Per Person-Month (THB)	Activity Labor Cost (THB)
Management	10%+(10%*40%*2) = 18%	12*0.18 = 2.16	90,000	194,400
Analysis	10%+(5%*40%*2) = 14%	12*0.14 = 1.68	50,000	84,000
Design	15%*40%*2 = 12%	12*0.12 = 1.44	50,000	72,000
Implementation	50%*40%*2 = 40%	12*0.4 = 4.8	30,000	144,000
Testing	20%*40%*2 = 16%	12*0.16 = 1.92	30,000	57,600
			Total Project Labor Cost	552,000

Part B Work Breakdown Structure (WBS) [Estimated time: 30 minutes]

Create a work breakdown structure (WBS), with activities and dependencies, for your term project in the Software Development Practice II course. You may apply problem decomposition and/or process decomposition techniques. See Software Project Management lecture slides, page 71-73. (This is an open-ended question. You may decompose the scope of work the way you like but make it reasonable. Do NOT copy from Part C, question 2)

Your project name is

Level 1 (WBS item)	Level 2 (Activity)	Dependency

Part C Project Schedule and Gantt Chart [Estimated time: 1 hour]

Learn how to use ProjectLibre to create a Gantt chart to present a WBS, activities, and schedule of a project.

Download and install **ProjectLibre open source** for both Windows and Mac from
https://www.projectlibre.com/product/1-alternative-microsoft-project-open-source
Tutorial: https://www.youtube.com/watch?v=9xwR4JCBaIU

Follow the instructions below step by step to explore a few capabilities of a typical project management tool. (Note that the tool calls WBS items and activities "tasks".) It works very much like a spreadheet.

New project

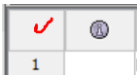
1. New, Project name = Test, Manager = your name, Start Date = 3 April 2024, leave Forward scheduled ticked, OK. The project is shown in Gantt view. The project uses the standard calendar, i.e. workdays are Mon-Fri, 8 hours per day.

2. Have a look at tab File, Task, Resource, View.

3. Create task Move to temporary room, Duration = 1 (or 1d) ("1 day?" in column Duration is only a placeholder. You need to type in 1 (or 1d).)

4. Select task Move to temporary room and right click > New to insert task Renovation as the first task (or select Task > Insert).

5. Select task Move to temporary room and right click > Indent to indent task as a subtask of the task Renovation (or select Task > Indent). See Duration of task Renovation. The task Renovation is now a "summary task".

6. Right click at the corner to explore task table. 

7. Right click at column Duration to insert/hide column Work. You can drag a column to move it around.

Dependency, network diagram, and critical path

1. New, Project name = Mor Prom, Manager = your name, Start Date = 3 April 2024, leave Forward scheduled ticked.

2. Create task, duration, and predecessor. (You may copy the task, duration, and predecessor columns to the tool. You may have to move up the column Predecessor to the left)

No	Task	Duration (day)	Predecessor	Remark
1	**Mor Prom project**			**Summary task level 1**
2	**Planning**			**Summary task level 2**
3	Create plan	6d		
4	**Analysis and design**			**Summary task level 2**
5	Analyze requirements	5d	3fs	Task 3 finishes to start task 5
6	Create prototype	4d	5ss	Task 5 starts to start task 6
7	Create design	5d	5fs;6fs	Task 5 finishes to start task 7, and Task 6 finishes to start task 7
8	**Procurement**			**Summary task level 2**
9	Acquire servers and software license	4d	7fs	
10	**Implementation**			**Summary task level 2**
11	Code	20d	9fs	
12	Test	4d	11fs	
13	Deploy	1d	12ff	Task 12 finishes to finish task 13
14	**Delivery**			**Summary task level 2**
15	Accept product	1d	13fs	
16	Take over by operation team	1d	15fs	
17	Deliver final document	1d	16sf	Task 16 starts to finish task 17
18	Issue a press release	1d	17fs	

The column Predecessor defines the dependencies between tasks. The dependencies are

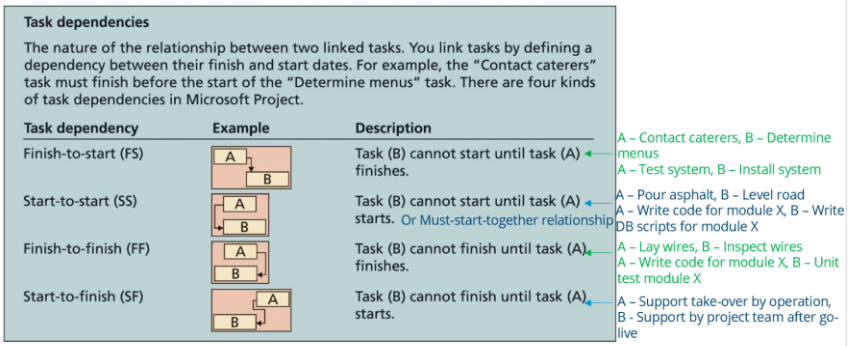


FIGURE 6-3 Task dependency types

3. See critical tasks (in red) and non-critical tasks (in blue).

4. Right click on bar chart area > Bar Styles > Total Slack. See shaded bar of total slack of non-critical task. (Total slack is the amount of time that the task can be delayed without affecting the project.)

5. Task > Zoom In to zoom in task calendar from day to hour of day.

6. Task > Zoom Out to zoom out task calendar from day to every 3 days, week, quarter, half year, year (4 quarters), and year (2 halves).

7. View > Network to see a network diagram.

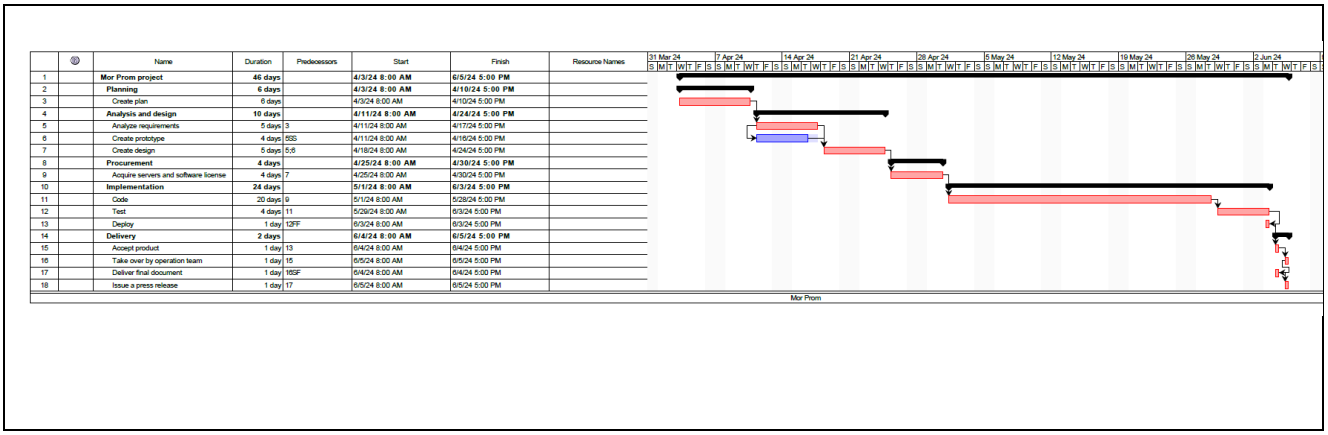
8. View > WBS to see a WBS.

9. View > Task Usage to see work (no. of hours) for each task.

10. File > Save as the project (.pod) to a folder.

11. File > Preview, Printer ProjectLibre PDF, Paper One big page, and click PDF button to save the project (.pdf) to a folder.

12. Crop the picture of your Gantt chart from the .pdf file and paste it here.



		Name	Duration	Predecessors	Start	Finish	Resource Names
1		**Mor Prom project**	**46 days**		**4/3/24 8:00 AM**	**6/5/24 5:00 PM**	
2		**Planning**	**6 days**		**4/3/24 8:00 AM**	**4/10/24 5:00 PM**	
3		Create plan	6 days		4/3/24 8:00 AM	4/10/24 5:00 PM	
4		**Analysis and design**	**10 days**		**4/11/24 8:00 AM**	**4/24/24 5:00 PM**	
5		Analyze requirements	5 days	3	4/11/24 8:00 AM	4/17/24 5:00 PM	
6		Create prototype	4 days	5SS	4/11/24 8:00 AM	4/16/24 5:00 PM	
7		Create design	5 days	5;6	4/18/24 8:00 AM	4/24/24 5:00 PM	
8		**Procurement**	**4 days**		**4/25/24 8:00 AM**	**4/30/24 5:00 PM**	
9		Acquire servers and software license	4 days	7	4/25/24 8:00 AM	4/30/24 5:00 PM	
10		**Implementation**	**24 days**		**5/1/24 8:00 AM**	**6/3/24 5:00 PM**	
11		Code	20 days	9	5/1/24 8:00 AM	5/28/24 5:00 PM	
12		Test	4 days	11	5/29/24 8:00 AM	6/3/24 5:00 PM	
13		Deploy	1 day	12FF	6/3/24 8:00 AM	6/3/24 5:00 PM	
14		**Delivery**	**2 days**		**6/4/24 8:00 AM**	**6/5/24 5:00 PM**	
15		Accept product	1 day	13	6/4/24 8:00 AM	6/4/24 5:00 PM	
16		Take over by operation team	1 day	15	6/5/24 8:00 AM	6/5/24 5:00 PM	
17		Deliver final document	1 day	16SF	6/4/24 8:00 AM	6/4/24 5:00 PM	
18		Issue a press release	1 day	17	6/5/24 8:00 AM	6/5/24 5:00 PM	

Part D Risk Management [Estimated time: 30 minutes]

1. Suppose you are planning the Mor Prom project for Covid vaccine reservation and have some concerns about the project. Identify two negative risks for the project. Fill in all the details of the risk register. (See Software Project Management lecture slides, page 85-89.)

Risk ID	1
Rank	1
Risk Event	Project delay
Description	The project cannot finish by the planned date.
Risk Category	Process risk (schedule risk)
Root cause	The time is so tight as the app is required urgently for the Covid situation.
Triggers	Several project tasks are behind schedule.
Potential responses	Request for a few more competent team members. (risk escalation)
Risk owner	Project manager
Probability	High
Impact	High
Status	The project manager is asking for support from the project sponsor.

Risk ID	2
Rank	2
Risk Event	Limited accessibility of the app
Description	The app is not accessible to all Thai people for Covid vaccine reservation.
Risk Category	People risk
Root cause	Some people do not have mobile phones. Some are computer-illiterate.
Triggers	Some concerns have been raised by senior citizens.
Potential responses	- Introduce a feature that allows a user to make vaccine reservation for others. (risk mitigation) - Inform the project sponsor that other options for vaccine reservation must be provided. (risk escalation)
Risk owner	Project manager and project sponsor (Minister of Public Health)

Probability	High
Impact	Medium
Status	- The project manager, the project sponsor, and the team have discussed and agreed to include the additonal feature in the plan despite the tight schedule. - The project sponsor will provide other ways to reserve the vaccine, such as house visit by local staff in all areas.

Submit your answer back to Assignment as a PDF file.